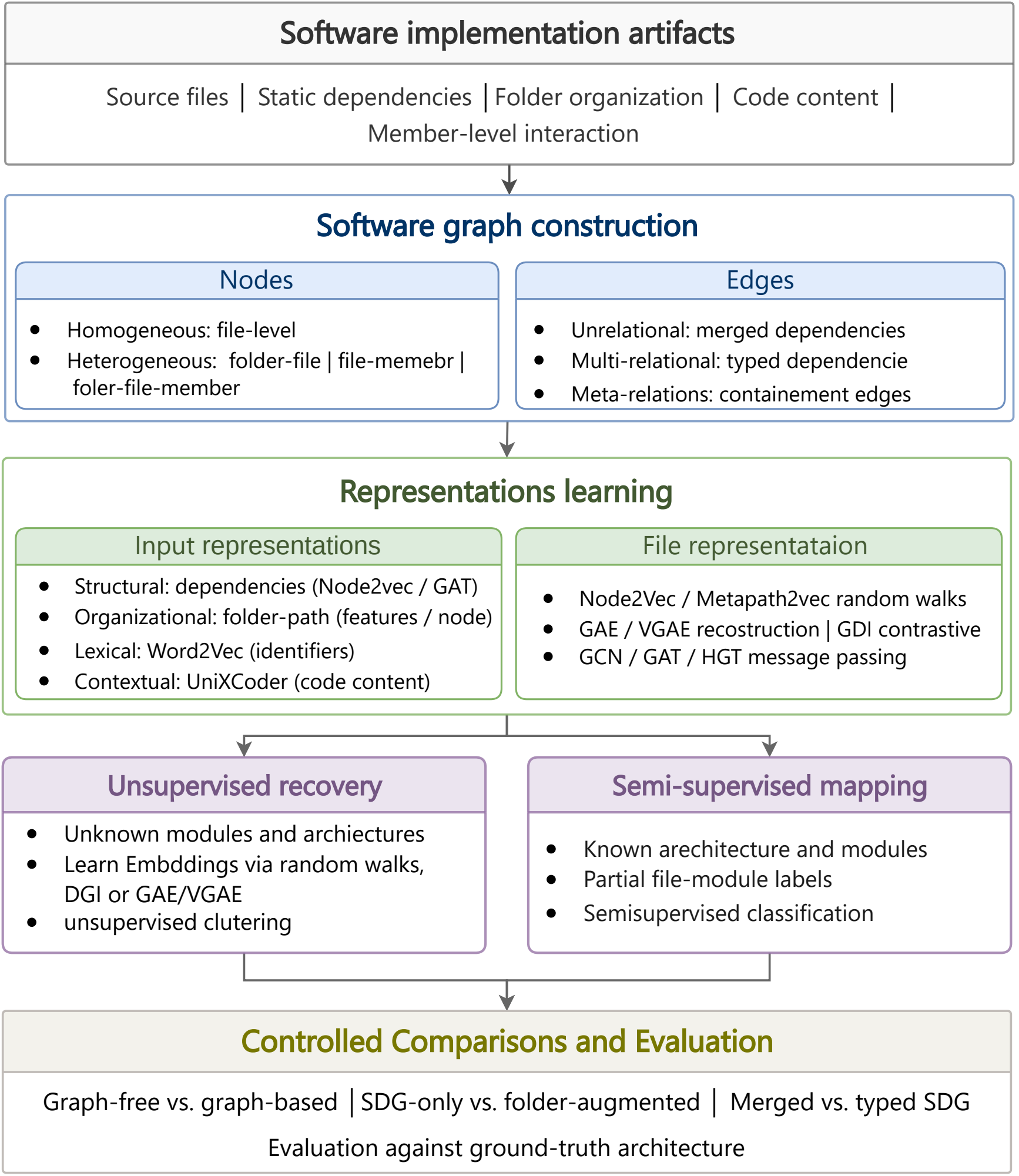

Software implementation artifacts
Source files | Static dependencies | Folder organization | Code content | Member-level interaction
Software graph construction
Nodes
Homogeneous: file-level
Heterogeneous: folder-file | file-memebr | foler-file-member
Edges
Unrelational: merged dependencies
Multi-relational: typed dependencie
Meta-relations: containement edges
Representations learning
Input representations
Structural: dependencies (Node2vec / GAT)
Organizational: folder-path (features / node)
Lexical: Word2Vec (identifiers)
Contextual: UniXCoder (code content)
File representataion
Node2Vec / Metapath2vec random walks
GAE / VGAE recostruction | GDI contrastive
GCN / GAT / HGT message passing
Unsupervised recovery
Unknown modules and archiectures
Learn Embddings via random walks, DGI or GAE/VGAE
unsupervised clutering
Semi-supervised mapping
Known arechitecture and modules
Partial file-module labels
Semisupervised classification
Controlled Comparisons and Evaluation
Graph-free vs. graph-based | SDG-only vs. folder-augmented | Merged vs. typed SDG
Evaluation against ground-truth architecture